Experiment 8

NMOS Linear Voltage Regulator using a 741 Op-Amp: 1.2 V Output from a 3 to 5 V Supply

Important: All numerical values in the Section 6 reference plots (V_TH, k_n, lambda, 741 gain and offset, output swing) are illustrative only, chosen to show the shape of the curves. Your power MOSFET and your 741 will have different values, and a real 741 will show far larger regulation errors than the simulation, because the measurement is limited by the reference stability, the op-amp offset and drift, and the resolution of your multimeter. Take your own V_TH, V_GS(max), R_DS(on) and current ratings from your MOSFET datasheet, and do not copy the numbers from Section 6 into your report.

1. Objective

To build a linear voltage regulator in which a 741 op-amp drives the gate of an NMOS power transistor connected as a source follower, so that the output is held at 1.2 V while the input supply varies from 3 V to 5 V. To measure line regulation, load regulation and dropout voltage, and to explain how negative feedback makes the output follow the reference. To measure the efficiency of the regulator and see why a switching regulator, the subject of the next experiment, can do much better.

2. Background

The 741 compares the reference V_REF (non-inverting input) with the output V_OUT (inverting input) and drives the MOSFET gate. The source of the MOSFET is the output, so any fall in V_OUT increases V_GS and the MOSFET supplies more current, and any rise in V_OUT reduces V_GS. With feedback taken directly from the output, the loop forces:

$$V_OUT = V_REF \quad (error = V_REF / (1 + loop\ gain))$$

Small-signal analysis of the source follower inside the loop, with open-loop op-amp gain A, MOSFET transconductance g_m and output resistance r_o, gives the line sensitivity and closed-loop output resistance:

$$dV_OUT / dV_IN = 1 / (r_o\ g_m\ (1 + A)) \qquad R_OUT = 1 / (g_m\ (1 + A) + 1/r_o)$$

Both are very small because A is about 2 x 10^5 for a 741, which is why a good source-follower regulator shows only microvolt-level changes in theory. In the lab these are hidden by reference drift, op-amp offset and meter resolution.

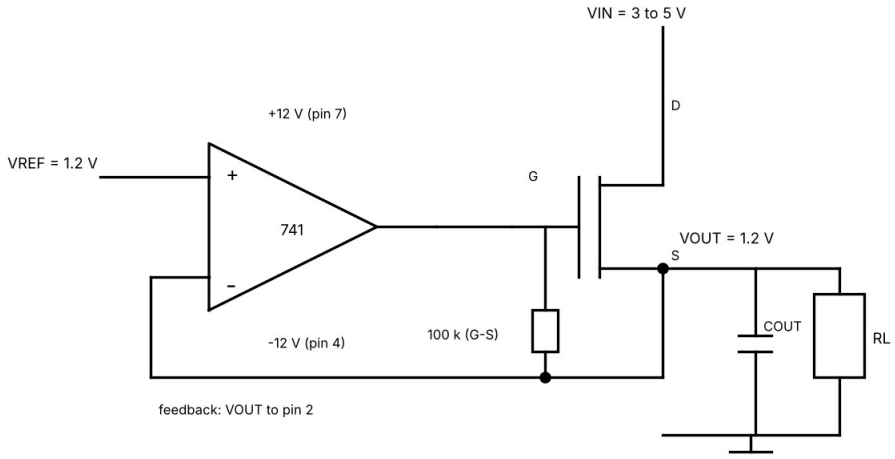
Why the op-amp needs its own supply. The MOSFET gate must sit at V_OUT + V_GS, which is about 4.2 to 4.6 V for a typical power MOSFET at 1.2 V output. A 741 cannot swing its output to within about 2 V of its supply rails, and its input common-mode range also stops about 2 V short of each rail. Powered from the 3 to 5 V input, it could neither drive the gate high enough nor accept a 1.2 V input. The op-amp is

therefore powered from separate +12 V and -12 V supplies, while only the MOSFET drain is connected to the 3 to 5 V input.

Why a minimum load is needed. A source follower can supply current but cannot sink it. If the load current is zero, the output has no path to discharge and V_OUT can sit above V_REF. The permanent 1 k Ohm bleeder sets a minimum load of about 1.2 mA.

Dropout voltage. When V_IN falls to near 1.2 V, the op-amp output saturates at its maximum swing and the MOSFET is fully on. The dropout is then approximately I_LOAD x R_DS(on), plus the headroom that the datasheet gives at your gate drive.

3. Circuit and Components



Item	Value	Remarks
Error amplifier	741 op-amp (DIP-8)	Pin 7 = +12 V, pin 4 = -12 V, pin 3 (+) = V_REF, pin 2 (-) = V_OUT, pin 6 = output
Pass transistor	Power N-MOSFET	Lab-supplied power MOSFET; drain to V_IN, source is V_OUT. Note its V_TH and V_GS(max)
Gate-source resistor	100 k Ohm	Keeps the MOSFET off when the op-amp is not powered
Bleeder load	1 k Ohm	Always connected. Gives about 1.2 mA minimum load (see Section 2)
Load resistors, R_L	120, 47, 22, 12 Ohm	0.5 W or higher; switched in parallel with the bleeder to give roughly 10, 25, 55, 100 mA
Output capacitor, C_OUT	10 uF	Across V_OUT to ground, close to the load
Reference	2.5 V shunt reference (TL431 as a diode) with 4.7 k Ohm from +12 V; 10 k Ohm pot across it	Wiper gives V_REF = 1.200 V. A 5.1 V zener with a 10 k Ohm pot may be substituted

Supplies	V_IN: 3 to 5 V; V_AUX: +12 V and -12 V	Three independent outputs with common ground; V_IN current limit set to 200 mA

4. Safety Limits and Precautions

- Check your MOSFET datasheet for V_GS(max). With +12 V op-amp supply the gate can reach about +10 V. If V_GS(max) is below about 12 V (some logic-level devices are rated at only 10 V), reduce the op-amp supplies to +/-8 V or add a gate-to-source zener, before powering up.

- Set the V_IN supply current limit to 200 mA and never exceed 5 V. The operating load current must stay at or below 100 mA.

- Worst-case dissipation in the MOSFET is (5 V - 1.2 V) x 0.1 A, about 0.4 W, which a TO-220 package handles without a heat sink. The case still warms up, so do not touch it during measurement.

- Power-up order: switch on the +12 V and -12 V supplies first, confirm the pins of the 741, then apply V_IN. Switch off in the reverse order.

- Set V_REF to 1.200 V with the multimeter before connecting it to pin 3.

- When measuring I_IN or the control-supply currents with the multimeter, insert it in series on its mA range only. Return it to the voltage range and remove it from the circuit before changing V_IN or the load.

- Keep the AC test or scope probe ground on the circuit ground only. Do not short V_OUT to ground.

5. Procedure

Part A - Set-up and check. Build the circuit of Section 3 with only the bleeder as load. Apply +12 V and -12 V, set V_REF to 1.200 V, then raise V_IN to 3.0 V. Measure V_OUT and the gate voltage V_G. V_OUT should equal V_REF to within a few millivolts, and V_G should equal V_OUT plus the V_GS of your MOSFET.

Part B - Line regulation. Connect R_L = 22 Ohm in parallel with the bleeder. Vary V_IN from 3.0 V to 5.0 V in steps of 0.25 V and record V_OUT and V_G in Table 1. Calculate line regulation as (V_OUT,max - V_OUT,min) / (5 V - 3 V) in mV/V.

Part C - Load regulation. At V_IN = 3 V and again at V_IN = 5 V, step the load through the bleeder only, 120, 47, 22 and 12 Ohm (about 1, 10, 25, 55 and 100 mA). Measure the actual load current with the multimeter or from V_OUT / R_L using the measured resistor value, and record V_OUT and V_G in Table 2. Calculate load regulation as the change in V_OUT divided by the change in load current (mV/A); this is the closed-loop output resistance.

Part D - Dropout. With R_L = 12 Ohm connected, reduce V_IN from 3.0 V downward in steps of 0.2 V and record V_OUT in Table 3. Define the dropout voltage as the V_IN at which V_OUT has fallen 1% (12 mV) below its regulated value.

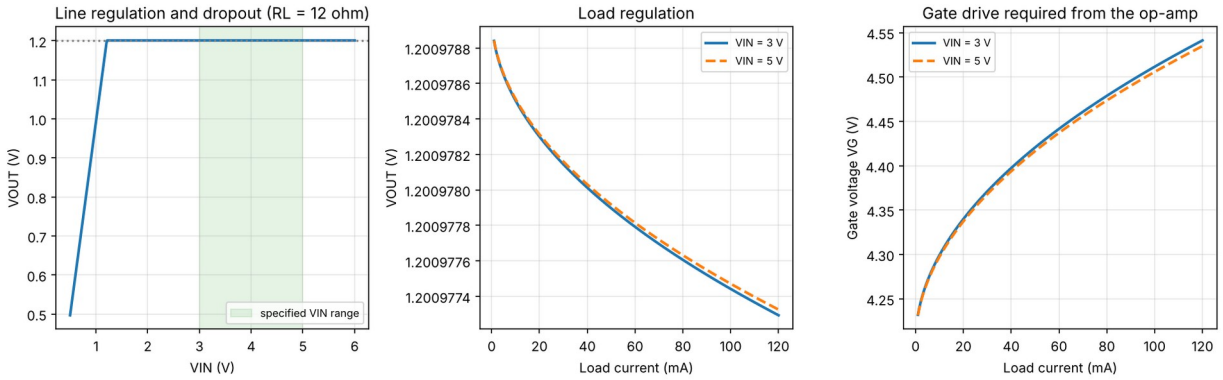
Part E - Efficiency. With R_L = 12 Ohm connected (about 100 mA), set V_IN to 3.0, 3.5, 4.0, 4.5 and 5.0 V. At each setting record the input current I_IN drawn from the V_IN supply (from the supply display or a multimeter in series) and the load current

I_LOAD, and fill in Table 4. Calculate P_IN = V_IN x I_IN, P_OUT = V_OUT x I_LOAD, the pass-only efficiency eta = P_OUT / P_IN, and the heat dissipated in the MOSFET P_diss = P_IN - P_OUT. Then read the currents drawn from the +12 V and -12 V supplies once, compute P_AUX = 12 V x (I_AUX+ + I_AUX-), and calculate the total efficiency P_OUT / (P_IN + P_AUX) at V_IN = 3 V and 5 V.

Part F - Stability check. At 100 mA load, view V_OUT on the oscilloscope with AC coupling. A clean trace with only small noise means the loop is stable. The 741 output drives the gate directly, so no gate resistor is used. If you do see oscillation, increase C_OUT and record what changes.

6. Reference Simulation (illustrative values only)

The figure shows a numerical solution of the regulator for illustrative device values (V_TH = 3 V, k_n = 2 A/V^2, lambda = 0.02 /V, A = 2 x 10^5, op-amp offset 1 mV, output swing limited to +/-10 V). Left: V_OUT against V_IN for R_L = 12 Ohm, showing dropout below about 1.2 V and a flat regulated output across the 3 to 5 V range. Middle: V_OUT against load current at V_IN = 3 V and 5 V, where the change is only a few microvolts in this ideal model. Right: the gate voltage the op-amp must supply, which rises slowly with load current and always stays well below the 741 positive swing. Your measured curves will show larger variations, and the V_G values will depend on your own MOSFET.



7. Efficiency and the Case for a Switching Regulator

The load current of a linear regulator flows from the input, through the pass transistor, to the output. The input current is therefore almost equal to the load current, and the efficiency is simply:

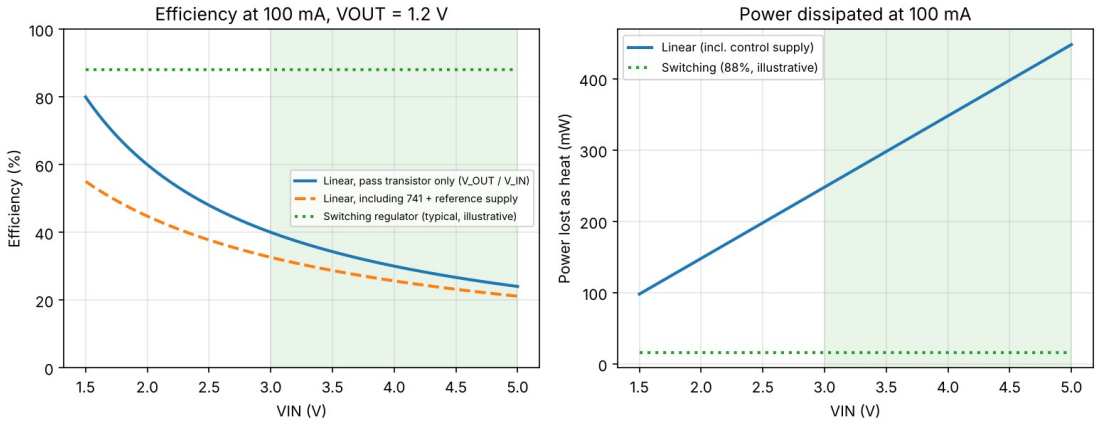
$$eta = P_OUT / P_IN = (V_OUT \times I) / (V_IN \times I) = V_OUT / V_IN$$

The power (V_IN - V_OUT) x I is turned into heat in the MOSFET, whatever the quality of the op-amp or the reference. For 1.2 V from 5 V the best possible efficiency is 24%, so three quarters of the input power is wasted, and from 3 V it is 40%. The 741 and the reference also draw their own current from the +12 V and -12 V supplies, which lowers the total efficiency further. The only ways to improve a linear regulator are to lower V_IN or to raise V_OUT.

A switching regulator avoids this by never operating the series transistor in its linear region. The transistor is either fully on, so V_DS is only I x R_DS(on) and the product V_DS x I is very small, or fully off, so the current is zero. An inductor and a capacitor store and pass energy between the switch pulses and average the pulse train to a DC output. For a step-down (buck) converter in continuous conduction:

$$V_OUT = D \times V_IN, \quad D = t_on / T$$

To get 1.2 V from 3 to 5 V the duty cycle is about 0.4 down to 0.24. The remaining losses are conduction loss I^2 R_DS(on), switching loss in the transistor during the turn-on and turn-off transitions, gate-drive power, the inductor winding and core loss, and the diode drop (or the second switch). They give typical efficiencies of 80 to 95% that stay nearly flat as V_IN changes. The main costs are output ripple at the switching frequency, electromagnetic interference, extra parts (an inductor and a diode or second switch), and a slower transient response.



The figure above uses illustrative values (100 mA load, 70 mW for the control supplies, 88% for the switching regulator). The efficiency you measure in Part E is the baseline that your switching regulator will have to beat. A summary of the trade-offs follows.

Property	Linear regulator (this experiment)	Switching regulator (next experiment)
Efficiency	About V_OUT / V_IN (24 to 40% here), falls as V_IN rises	Typically 80 to 95%, almost independent of V_IN
Heat in the series device	(V_IN - V_OUT) x I, needs a heat sink at higher current	Small: only conduction and switching losses
Output voltage range	Step-down only (V_OUT below V_IN)	Buck, boost or inverting from the same parts
Output noise	Very low, no switching noise	Ripple at switching frequency, EMI, needs filtering
Parts and size	Few parts, no magnetics	Inductor, diode or second switch, capacitors, controller
Transient response	Fast, wide loop bandwidth	Limited by the LC filter and control loop
Battery life	Poor when V_IN is much larger than V_OUT	Much longer for the same battery

8. Observation Tables (fill in using your own devices)

Table 1: Line regulation at R_L = 22 Ohm (with bleeder).

V_IN (V)	V_OUT (V)	V_G (V)
3.00		
3.25		
3.50		
3.75		
4.00		
4.25		
4.50		
4.75		
5.00		

Table 2: Load regulation.

R_L (nominal)	I_LOAD (mA)	V_OUT (V) at V_IN = 3 V	V_OUT (V) at V_IN = 5 V	V_G (V) at V_IN = 3 V
Bleeder only				
120 Ohm				
47 Ohm				
22 Ohm				
12 Ohm				

Table 3: Dropout at R_L = 12 Ohm.

V_IN (V), R_L = 12 Ohm	V_OUT (V)

Table 4: Efficiency at R_L = 12 Ohm (about 100 mA).

V_IN (V)	I_LOAD (mA)	I_IN (mA)	P_IN (mW)	P_OUT (mW)	eta, pass only (%)	P_diss (mW)
3.0						
3.5						
4.0						
4.5						
5.0						

Control-supply readings (record once): I_AUX+ = _____ mA, I_AUX- = _____ mA, P_AUX = _____ mW. Total efficiency at V_IN = 3 V: _____ %, at V_IN = 5 V: _____ %.

9. Analysis and Discussion Questions

- What line regulation (mV/V) and load regulation (mV/A) did you measure? Compare them with the ideal result of Section 2 and identify what limits your measurement.
- What was the V_G you measured at 100 mA? Compare V_G - V_OUT with the threshold voltage on your MOSFET datasheet.
- What dropout voltage did you measure at 100 mA? Relate it to R_DS(on) of your MOSFET and the 741 output swing.
- Why can the regulator not be powered with the op-amp on the same 3 to 5 V supply as the drain? What alternative op-amp type would allow it?
- What happens to V_OUT if the bleeder is removed and the load is disconnected? Explain.
- How would you change the circuit to obtain a regulated output of 2.4 V with the same V_REF? Give the resistor ratio and note what happens to the dropout.
- What is the power dissipated in the MOSFET at V_IN = 5 V and 100 mA, and how does the efficiency compare with V_IN = 3 V?
- Why is the efficiency of this regulator limited to V_OUT / V_IN, however well the op-amp is designed? What would the efficiency be if V_IN were 12 V?
- By how much does the control supply (741 and reference) lower the total efficiency? Which part of it would remain in a switching regulator?
- In a switching regulator the series transistor is either fully on or fully off. Explain why this makes the loss small, and name two loss mechanisms that remain.
- A buck converter delivers 120 mW at 88% efficiency. How much heat does it dissipate, and how does this compare with your measured P_diss at V_IN = 5 V?